| Symbol | Unit | Meaning |
|---|---|---|
| *Indices* | | |
| $i,\ j$ | Token | Input (sold) and output (bought) endpoint tokens; the ordered pair is $i \to j$. |
| $k$ | Token | Candidate vehicle token used as the route intermediate; $k \notin \{i, j\}$. |
| $\ell,\ p$ | Token / pool | $\ell$ indexes tokens in cross-token sums; $p$ indexes liquidity pools. |
| $t,\ w$ | UTC day / UTC week | $t$ indexes calendar days; $w$ indexes calendar weeks in settlement variables. |
| $q$ | USD | Input quote notional. Un-suffixed route-cost columns use $q = \$10{,}000$. |
| $r$ | Route unit | Reconstructed source-to-sink route unit. A coherent $i \to k \to j$ component contributes one $r$ regardless of its number of legs; a split or join contributes one $r$ per reconstructed source-sink pair. |
| *Route and liquidity aggregates* | | |
| $\mathrm{Vol}_t,\ \mathrm{DVol}_t$ | USD | $\mathrm{Vol}_t$ is total realized USD volume across all direct and indirect route units on day $t$. $\mathrm{DVol}_t$ restricts it to direct routes, so $0 \le \mathrm{DVol}_t \le \mathrm{Vol}_t$. |
| $\mathrm{IVol}_t,\ \mathrm{IVol}_{k,t}$ | USD | $\mathrm{IVol}_t$ restricts $\mathrm{Vol}_t$ to indirect routes, so $0 \le \mathrm{IVol}_t \le \mathrm{Vol}_t$. $\mathrm{IVol}_{k,t}$ further restricts that volume to routes using vehicle $k$, so $0 \le \mathrm{IVol}_{k,t} \le \mathrm{IVol}_t$. |
| $N_t,\ N^{\mathrm{src}}_{k,t},\ N^{\mathrm{sink}}_{k,t}$ | Route-unit count | $N_t$ counts all reconstructed route units $r$ on day $t$, not their individual legs. $N^{\mathrm{src}}_{k,t}$ and $N^{\mathrm{sink}}_{k,t}$ restrict that count to source and sink roles for $k$; each lies between 0 and $N_t$. |
| $N^I_t,\ N^I_{k,t}$ | Route-unit count | $N^I_t$ restricts $N_t$ to route units with at least one intermediate token. $N^I_{k,t}$ further restricts that count to units with intermediate $k$, so $0 \le N^I_{k,t} \le N^I_t \le N_t$; superscript $I$ denotes indirect routes. |
| $\mathcal{A}_t,\ \mathcal{A}^k_t,\ \mathcal{M}^k_t$ | Sets of token pairs | $\mathcal{A}_t$ is the set of endpoint pairs active in indirect routes on day $t$. $\mathcal{A}^k_t$ restricts it to pairs using $k$, and $\mathcal{M}^k_t$ further restricts it to pairs for which $k$ has the largest candidate-vehicle volume; $\mathcal{M}^k_t \subseteq \mathcal{A}^k_t \subseteq \mathcal{A}_t$. |
| $\mathrm{Vol}^{\mathrm{in}}_{k,t},\ \mathrm{Vol}^{\mathrm{out}}_{k,t}$ | USD | $\mathrm{Vol}^{\mathrm{in}}_{k,t}$ and $\mathrm{Vol}^{\mathrm{out}}_{k,t}$ are inbound and outbound route-leg volumes assigned to token $k$ on day $t$. |
| $\mathrm{Vol}^{\mathrm{src}}_{k,t},\ \mathrm{Vol}^{\mathrm{sink}}_{k,t}$ | USD | Source- and sink-role restrictions of $\mathrm{Vol}_t$ for token $k$: $0 \le \mathrm{Vol}^{\mathrm{src}}_{k,t} \le \mathrm{Vol}_t$ and $0 \le \mathrm{Vol}^{\mathrm{sink}}_{k,t} \le \mathrm{Vol}_t$. |
| $\mathcal{K}$ | Set of tokens | Prespecified candidate set $\{\mathrm{WETH}, \mathrm{USDC}, \mathrm{USDT}, \mathrm{DAI}, \mathrm{WBTC}\}$. |
| $\mathcal{L}_{k,t},\ m_p$ | Set of pools / token count | $\mathcal{L}_{k,t}$ contains Uniswap V3 daily-snapshot pools with candidate $k$ on one side, exact token contracts identified in the persisted V3 swap archive, and $0 < \mathrm{TVL}_{p,t} \le 10$ billion USD. $m_p$ is the number of pool tokens in $\mathcal{K}$, so $m_p \in \{1, 2\}$. |
| $\mathrm{TVL}_{p,t},\ L_{k,t}$ | USD | $\mathrm{TVL}_{p,t}$ is pool $p$'s day-$t$ USD TVL from the V3 daily snapshot. $L_{k,t} = \sum_{p \in \mathcal{L}_{k,t}} \mathrm{TVL}_{p,t}/m_p$; a one-candidate pool contributes all TVL to that candidate and a two-candidate pool contributes one half to each. |
| $R^{\mathrm{WETH}}_t$ | Daily log-return fraction | Day-$t$ log return of the WETH price used to construct downside stress. |
| *Route-cost quote objects* | | |
| $\mathcal{P}_{k,t,q}$ | Set of token pairs | Ordered endpoint pairs eligible for $k$ and quoted on day $t$ at notional $q$. |
| $\mathcal{D}_{k,t,q},\ \mathcal{I}_{k,t,q}$ | Sets of token pairs | $\mathcal{D}_{k,t,q} \subseteq \mathcal{P}_{k,t,q}$ and $\mathcal{I}_{k,t,q} \subseteq \mathcal{P}_{k,t,q}$ restrict the broad pair set to pairs with an executable direct route or an executable indirect route via $k$. |
| $\mathcal{C}_{k,t,q}$ | Set of token pairs | Common-support pairs for candidate $k$, day $t$, and notional $q$: $\mathcal{C}_{k,t,q} = \mathcal{D}_{k,t,q} \cap \mathcal{I}_{k,t,q}$, so both the direct and indirect routes are executable. |
| $\mathcal{T}_{k,t,q},\ \mathcal{W}_{k,t,q}$ | Sets of token pairs | $\mathcal{T}_{k,t,q} \subseteq \mathcal{D}_{k,t,q}$ is the thin-direct subset. $\mathcal{W}_{k,t,q} \subseteq \mathcal{C}_{k,t,q}$ contains common-support pairs for which the indirect route beats the direct route, equivalently $\Delta C^D_{i,j,k,q,t} < 0$. |
| $D_{i,j,q,t},\ I_{i,j,k,q,t},\ T_{i,j,q,t}$ | Indicator (0/1) | Indicators for direct-route availability ($D$), indirect-route-through-$k$ availability ($I$), and an executable direct route returning less than $0.9q$ ($T$). |
| $O^D_{i,j,q,t},\ O^I_{i,j,k,q,t}$ | USD | Quoted output values; superscripts $D$ and $I$ denote direct and indirect routes. |
| $\Delta C^D_{i,j,k,q,t}$ | Fraction | Pair-level direct cost advantage $(O^D_{i,j,q,t} - O^I_{i,j,k,q,t})/O^D_{i,j,q,t}$ on $\mathcal{C}_{k,t,q}$; positive values favor the direct route. |
| *Settlement objects and operators* | | |
| $\mathcal{R}_{k,w}$ | Set of route units | Receipt-audited matched route units using vehicle $k$ in UTC week $w$. |
| $\mathcal{R}^{\mathrm{transfer}}_{k,w}$ | Set of route units | Members whose receipt logs a transfer of vehicle $k$; $\mathcal{R}^{\mathrm{transfer}}_{k,w} \subseteq \mathcal{R}_{k,w}$. |
| $\lvert\mathcal{S}\rvert,\ \mathbf{1}_{\{\cdot\}}$ | Count / indicator (0/1) | Cardinality of any finite set $\mathcal{S}$ and an indicator equal to one when its subscripted condition is true. |
| $\Delta_7$ | 7 days | Seven-day forward change in any daily variable $X_t$: $\Delta_7 X_t = X_{t+7} - X_t$. |

| Variable | Formula | Unit | Data column | Definition |
|---|---|---|---|---|
| *Vehicle-use measures* | | | | |
| $\mathrm{VehicleShare}_{k,t}$ | $\dfrac{\mathrm{IVol}_{k,t}}{\mathrm{IVol}_t}$ | Fraction (0–1) | `bridge_share` | Fraction of day-$t$ indirect-route USD volume routed through $k$. |
| $\mathrm{AllRouteVehicleShare}_{k,t}$ | $\dfrac{\mathrm{IVol}_{k,t}}{\mathrm{Vol}_t}$ | Fraction (0–1) | `all_route_bridge_share` | Fraction of day-$t$ all-route USD volume routed indirectly through $k$. |
| $\mathrm{VehicleCountShare}_{k,t}$ | $\dfrac{N^I_{k,t}}{N^I_t}$ | Fraction (0–1) | `bridge_count_share` | Fraction of day-$t$ indirect route units that use $k$. |
| $\mathrm{PairCoverage}_{k,t}$ | $\dfrac{\lvert\mathcal{A}^k_t\rvert}{\lvert\mathcal{A}_t\rvert}$ | Fraction (0–1) | `pair_coverage` | Fraction of active indirect-route endpoint pairs that use $k$. |
| $\mathrm{MainVehiclePairShare}_{k,t}$ | $\dfrac{\lvert\mathcal{M}^k_t\rvert}{\lvert\mathcal{A}_t\rvert}$ | Fraction (0–1) | `pair_main_vehicle_share` | Fraction of active indirect-route endpoint pairs for which $k$ has the largest candidate-vehicle volume. |
| $\mathrm{IVol}_{k,t}$ | | USD | `bridge_volume_usd` | Sum of realized USD volume over indirect routes using vehicle $k$ on day $t$. |
| *Network and route-denominator controls* | | | | |
| $\mathrm{VolShare}_{k,t}$ | $\dfrac{\mathrm{Vol}^{\mathrm{in}}_{k,t} + \mathrm{Vol}^{\mathrm{out}}_{k,t}}{\sum_\ell (\mathrm{Vol}^{\mathrm{in}}_{\ell,t} + \mathrm{Vol}^{\mathrm{out}}_{\ell,t})}$ | Fraction (0–1) | `vol_share` | $\mathrm{Vol}^{\mathrm{in}}_{k,t}$ and $\mathrm{Vol}^{\mathrm{out}}_{k,t}$ are inbound and outbound route-leg USD volumes for $k$; $\ell$ indexes every token in the day-$t$ network. |
| $\mathrm{Betweenness}_{k,t}$ | $\dfrac{N^I_{k,t}}{N_t - N^{\mathrm{src}}_{k,t} - N^{\mathrm{sink}}_{k,t}}$ | Fraction (0–1) | `betweenness_centrality` | Fraction of day-$t$ intent routes on which $k$ is intermediate, excluding routes where $k$ is the source or sink. Superscripts identify each route role. |
| $\mathrm{Betweenness}^{\mathrm{vol}}_{k,t}$ | $\dfrac{\mathrm{IVol}_{k,t}}{\mathrm{Vol}_t - \mathrm{Vol}^{\mathrm{src}}_{k,t} - \mathrm{Vol}^{\mathrm{sink}}_{k,t}}$ | Fraction (0–1) | `volume_weighted_betweenness` | USD-volume analogue of $\mathrm{Betweenness}_{k,t}$ using the same route universe and source/sink exclusions. |
| $\mathrm{Vol}_t$ | $\mathrm{DVol}_t + \mathrm{IVol}_t$ | USD | `daily_all_route_volume_usd` | Total realized USD volume across direct and indirect route units on day $t$. |
| $\mathrm{IVol}_t$ | | USD | `daily_indirect_route_volume_usd` | Total realized USD volume across indirect route units on day $t$. |
| $\mathrm{IndirectRouteShare}_t$ | $\dfrac{\mathrm{IVol}_t}{\mathrm{Vol}_t}$ | Fraction (0–1) | `indirect_route_share` | Fraction of day-$t$ all-route USD volume executed through indirect routes. |
| *Liquidity measures* | | | | |
| $L_{k,t}$ | $\displaystyle\sum_{p \in \mathcal{L}_{k,t}} \frac{\mathrm{TVL}_{p,t}}{m_p}$ | USD | `vehicle_linked_liquidity_usd` | Candidate $k$'s allocated share of valid Uniswap V3 pool TVL: full TVL when $k$ is the pool's only candidate token and one half when both pool tokens are candidates. |
| $\mathrm{LPConc}_{k,t}$ | $\dfrac{L_{k,t}}{\sum_{\ell \in \mathcal{K}} L_{\ell,t}}$ | Fraction (0–1) | `lp_concentration` | Candidate $k$'s share of total vehicle-linked liquidity on day $t$. |
| $\mathrm{LogVehicleLiquidity}_{k,t}$ | $\ln(1 + L_{k,t})$ | Natural-log points | `log_vehicle_linked_liquidity` | Natural log of one plus $L_{k,t}$ measured in USD. |
| *Route-cost opportunity measures* | | | | |
| $\mathrm{DirectAvailable}_{k,t,q}$ | $\dfrac{\lvert\mathcal{D}_{k,t,q}\rvert}{\lvert\mathcal{P}_{k,t,q}\rvert}$ | Fraction (0–1) | `direct_available_share` | Fraction of pairs in $\mathcal{P}_{k,t,q}$ with $D_{i,j,q,t} = 1$. The un-suffixed data column fixes $q = \$10{,}000$. |
| $\mathrm{IndirectAvailable}_{k,t,q}$ | $\dfrac{\lvert\mathcal{I}_{k,t,q}\rvert}{\lvert\mathcal{P}_{k,t,q}\rvert}$ | Fraction (0–1) | `vehicle_available_share` | Fraction of pairs in $\mathcal{P}_{k,t,q}$ with $I_{i,j,k,q,t} = 1$, so both legs through $k$ are executable. |
| $\mathrm{IndirectOnlyAvailable}_{k,t,q}$ | $\dfrac{\lvert\mathcal{I}_{k,t,q} \setminus \mathcal{D}_{k,t,q}\rvert}{\lvert\mathcal{P}_{k,t,q}\rvert}$ | Fraction (0–1) | `no_direct_vehicle_available_share` | Fraction of pairs with $D_{i,j,q,t} = 0$ and $I_{i,j,k,q,t} = 1$: no executable direct route but an executable route through $k$. |
| $\mathrm{DirectDepth}_{k,t,q}$ | $\mathrm{median}_{\mathcal{D}_{k,t,q}} \left(O^D_{i,j,q,t}/q\right)$ | Output USD per input USD | `direct_depth_median` | Median direct-route output ratio $O^D_{i,j,q,t}/q$ over executable-direct pairs $(i, j) \in \mathcal{D}_{k,t,q}$. |
| $\mathrm{DirectCostAdvantage}_{k,t,q}$ | $\mathrm{median}_{\mathcal{C}_{k,t,q}} \Delta C^D_{i,j,k,q,t}$ | Fraction | `direct_cost_advantage_median` | Median $\Delta C^D_{i,j,k,q,t}$ over common-support pairs $(i, j) \in \mathcal{C}_{k,t,q}$; positive values favor the direct route and negative values favor the indirect route through $k$. |
| $\mathrm{IndirectBeatsDirect}_{k,t,q}$ | $\dfrac{\lvert\mathcal{W}_{k,t,q}\rvert}{\lvert\mathcal{C}_{k,t,q}\rvert}$ | Fraction (0–1) | `vehicle_beats_direct_share` | Fraction of $(i, j) \in \mathcal{C}_{k,t,q}$ for which $\Delta C^D_{i,j,k,q,t} < 0$, equivalently $O^I_{i,j,k,q,t} > O^D_{i,j,q,t}$. |
| $\mathrm{ThinDirectShare}_{k,t,q}$ | $\dfrac{\lvert\mathcal{T}_{k,t,q}\rvert}{\lvert\mathcal{P}_{k,t,q}\rvert}$ | Fraction (0–1) | `thin_direct_share` | Fraction of pairs with an executable direct quote but $O^D_{i,j,q,t}/q < 0.9$. This is a quote-quality proxy, not a direct measure of pool liquidity depth. |
| *Stress and dynamic variables* | | | | |
| $\mathrm{Stress}_t$ | $\max\{-R^{\mathrm{WETH}}_t, 0\}$ | Daily log-return fraction | `stress_downside` | Positive part of the negative day-$t$ WETH log return. |
| $\mathrm{StressEvent}_t$ | $\mathbf{1}_{\{\mathrm{Stress}_t \ge 0.08\}}$ | Indicator (0/1) | `stress_event_8pct` | Equals one when $\mathrm{Stress}_t$ is at least 0.08 on day $t$. |
| $\mathrm{VehicleShare}_{k,t+7}$ | | Fraction (0–1) | `future_bridge_share_t7` | Vehicle share for token $k$ seven calendar days after day $t$. |
| $\Delta_7 \mathrm{VehicleShare}_{k,t}$ | $\mathrm{VehicleShare}_{k,t+7} - \mathrm{VehicleShare}_{k,t}$ | Fraction points | `delta_bridge_share_t7` | Seven-day forward change for vehicle $k$, measured from day $t$. |
| *V4 settlement implementation measures* | | | | |
| $\mathrm{TransferIncidence}_{k,w}$ | $\dfrac{\lvert\mathcal{R}^{\mathrm{transfer}}_{k,w}\rvert}{\lvert\mathcal{R}_{k,w}\rvert}$ | Fraction (0–1) | `settlement_transfer_incidence` | Fraction of receipt-audited route units $r \in \mathcal{R}_{k,w}$ containing an ERC-20 Transfer log for intermediate vehicle $k$. |
| $\mathrm{ReceiptCount}_{k,w}$ | $\lvert\mathcal{R}_{k,w}\rvert$ | Route-unit count | `settlement_receipt_count` | Number of receipt-audited matched route units for vehicle $k$ in UTC week $w$. |